1.63 Shortest Distances in 3D: Planes

Worked Solutions

1)

The plane has normal vector $(1, 2, 9)$. The perpendicular from the origin is $\mathbf{r} = \lambda(1, 2, 9)$. Substitution gives $86\lambda = 344$, so $\lambda = 4$ and the foot is $(4, 8, 36)$. Thus the distance is

$$\sqrt{4^2 + 8^2 + 36^2} = 4\sqrt{86} \text{ units.}$$

Alternatively, the point-to-plane formula gives $\frac{|344|}{\sqrt{1^2+2^2+9^2}} = 4\sqrt{86}$ units.

2)

The plane normal is $(1, 1, 1)$. The perpendicular from $A = (2, 1, 2)$ is $\mathbf{r} = (2, 1, 2) + \lambda(1, 1, 1)$. Substitution into $x + y + z = 3$ gives $3\lambda + 5 = 3$, so $\lambda = -\frac{2}{3}$. The foot is $(\frac{4}{3}, \frac{1}{3}, \frac{4}{3})$ and

$$d = \sqrt{(-\tfrac{2}{3})^2 + (-\tfrac{2}{3})^2 + (-\tfrac{2}{3})^2} = \frac{2\sqrt{3}}{3} \text{ units.}$$

Alternatively, $d = \frac{|2+1+2-3|}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$ units.

3 a)

The plane normal is $(1, -1, -1)$.

The perpendicular from $P = (10, 3, 1)$ is $\mathbf{r} = (10, 3, 1) + \lambda(1, -1, -1)$. Substitution gives $3\lambda + 3 = 0$, so $\lambda = -1$ and the foot is $(9, 4, 2)$. Therefore

$$d = \sqrt{(-1)^2 + 1^2 + 1^2} = \sqrt{3} \text{ units.}$$

Alternatively, $d = \frac{|10-3-1-3|}{\sqrt{3}} = \sqrt{3}$ units.

3 b)

The plane normal is $(2, 5, -8)$. The perpendicular from $P = (-3, 2, 8)$ has form $\mathbf{r} = (-3, 2, 8) + \lambda(2, 5, -8)$. Substitution gives $93\lambda - 66 = 0$, so $\lambda = \frac{22}{31}$ and the foot is $(-\frac{49}{31}, \frac{172}{31}, \frac{72}{31})$. Hence

$$d = \frac{66}{\sqrt{93}} = \frac{22\sqrt{93}}{31} \text{ units.}$$

3 c)

The plane normal is $(-5, -3, 9)$. The perpendicular from $P = (-9, 0, 60)$ is $\mathbf{r} = (-9, 0, 60) + \lambda(-5, -3, 9)$. Substitution gives $115\lambda + 575 = 0$, so $\lambda = -5$ and the foot is $(16, 15, 15)$. Thus

$$d = \sqrt{25^2 + 15^2 + (-45)^2} = 5\sqrt{115} \text{ units.}$$

Alternatively, $d = \frac{575}{\sqrt{115}} = 5\sqrt{115}$ units.

3 d)

The plane normal is $(0, -1, -2)$. The perpendicular from $P = (6, 6, -1)$ is $\mathbf{r} = (6, 6, -1) + \lambda(0, -1, -2)$. Substitution gives $5\lambda - 5 = 0$, so $\lambda = 1$ and the foot is $(6, 5, -3)$. Therefore

$$d = \sqrt{0^2 + (-1)^2 + (-2)^2} = \sqrt{5} \text{ units.}$$

Alternatively, $d = \frac{|0-6+2-1|}{\sqrt{5}} = \sqrt{5}$ units.

4)

The plane normal is $(4, 1, -15)$. The perpendicular from $A = (-10, 20, -50)$ is $\mathbf{r} = (-10, 20, -50) + \lambda(4, 1, -15)$. Substitution gives $242\lambda - 968 = 0$, so $\lambda = 4$ and the foot is $(6, 24, -110)$. Hence

$$d = \sqrt{16^2 + 4^2 + (-60)^2} = 44\sqrt{2} \text{ units.}$$

Alternatively, $d = \frac{968}{\sqrt{242}} = 44\sqrt{2}$ units.

5 a)

Equating coordinates of the two lines gives

$$30\lambda + 80\mu = 10, \qquad \lambda + 3\mu = 0, \qquad 5\lambda - 5\mu = 20.$$

The first two equations give $\lambda = 3$ and $\mu = -1$, which satisfy the third. Substitution into l_1 gives $P = (100, 8, 5)$.

5 b)

The plane normal is $(1, -2, 3)$. For $P = (100, 8, 5)$,

$$|1(100) - 2(8) + 3(5) - 218| = 119, \qquad |(1, -2, 3)| = \sqrt{14}.$$

Thus the shortest distance is $\frac{119}{\sqrt{14}} = \frac{17\sqrt{14}}{2}$ units.

6)

The plane normal is $(3, -2, -14)$ and a point on the line has coordinates $(5 + 3\lambda, 28 + 5\lambda, 18 + \lambda)$. The distance condition gives

$$\frac{|3(5 + 3\lambda) - 2(28 + 5\lambda) - 14(18 + \lambda) + 39|}{\sqrt{209}} = \sqrt{209},$$

so $|-254 - 15\lambda| = 209$. Hence $\lambda = -3$ or $\lambda = -\frac{463}{15}$. The corresponding points are

$$Q = (-4, 13, 15) \quad \text{or} \quad Q = \left(-\frac{438}{5}, -\frac{379}{3}, -\frac{193}{15}\right).$$

Alternatively, the line through Q perpendicular to the plane has direction $(3, -2, -14)$. Substitution into the plane gives $\mu = (254 + 15\lambda)/209$. Setting the squared distance between Q and its foot equal to 209 gives

$$15\lambda^2 + 508\lambda + 1389 = 0 = (\lambda + 3)(15\lambda + 463),$$

which gives the same two points.

7 a)

The coal-bed plane has normal $(2, 1, 50)$. The perpendicular from $(25, 25, 0)$ is $(25, 25, 0) + \lambda(2, 1, 50)$. Substitution into the plane gives $2505\lambda + 5010 = 0$, so $\lambda = -2$ and the foot is $(21, 23, -100)$. The shortest distance is

$$\sqrt{(-4)^2 + (-2)^2 + (-100)^2} = 2\sqrt{2505} \text{ m.}$$

Alternatively, $d = \frac{5010}{\sqrt{2505}} = 2\sqrt{2505}$ m.

7 b)

Substitute the tunnel equation into $2x + y + 50z = -4935$:

$$2(25 + 10\lambda) + \left(25 + \frac{11}{7}\lambda\right) + 50\lambda = -4935,$$

so $\frac{501}{7}\lambda + 5010 = 0$ and $\lambda = -70$. The tunnel meets the coal bed at $(-675, -85, -70)$. The distance from the entrance is

$$\sqrt{(-700)^2 + (-110)^2 + (-70)^2} = 130\sqrt{30} \text{ m}.$$

8 a)

$\overrightarrow{BA} = (2, 1, 1)$ and $\overrightarrow{BC} = (2, 4, -1)$. Thus

$$\cos\theta = \frac{(2,1,1)\cdot(2,4,-1)}{\sqrt{6}\sqrt{21}} = \frac{7}{\sqrt{126}} = \frac{\sqrt{14}}{6},$$

and since θ is acute, $\sin\theta = \sqrt{1 - \frac{14}{36}} = \frac{\sqrt{22}}{6}$.

8 b)

A normal to the plane is

$$(2, 1, 1) \times (2, 4, -1) = (-5, 4, 6).$$

Using $A = (2, -1, 0)$ gives $-5x + 4y + 6z = -14$.

8 c)

Substituting $D = (3, 1, 5)$ gives $-5(3) + 4(1) + 6(5) = 19 \neq -14$, so D is not on the plane.

8 d)

The perpendicular from D has direction $(-5, 4, 6)$. The foot occurs at $\lambda = -\frac{3}{7}$ on $\mathbf{r} = (3, 1, 5) + \lambda(-5, 4, 6)$, giving $(\frac{36}{7}, -\frac{5}{7}, \frac{17}{7})$. Therefore

$$d = \frac{33}{\sqrt{77}} = \frac{3\sqrt{77}}{7} \text{ units}.$$

Alternatively, $d = \frac{|19-(-14)|}{\sqrt{77}} = \frac{3\sqrt{77}}{7}$ units.

8 e)

The area of ABC is

$$\frac{1}{2}|\overrightarrow{BA} \times \overrightarrow{BC}| = \frac{1}{2}\sqrt{(-5)^2 + 4^2 + 6^2} = \frac{\sqrt{77}}{2}.$$

Hence the tetrahedron volume is $\frac{1}{3} \cdot \frac{\sqrt{77}}{2} \cdot \frac{3\sqrt{77}}{7} = \frac{11}{2}$ units^3.

9 a)

The plane normal is $(1, 3, 2)$, so the perpendicular from the origin meets the plane at $\lambda(1, 3, 2)$. Substitution gives $14\lambda = 5$, and the distance is

$$\sqrt{\left(\frac{5}{14}\right)^2 + \left(\frac{15}{14}\right)^2 + \left(\frac{5}{7}\right)^2} = \frac{5}{\sqrt{14}} = \frac{5\sqrt{14}}{14}.$$

Alternatively, this is $\frac{|5|}{\sqrt{14}}$.

9 b)

Substitution of $A = (-13, 4, 3)$ gives $-13 + 3(4) + 2(3) = -13 + 12 + 6 = 5$, so A lies on the plane.

9 c)

The unit normal is $\hat{\mathbf{n}} = \frac{1}{\sqrt{14}}(1, 3, 2)$. Substitution of A gives $d = \frac{5}{\sqrt{14}} = \frac{5\sqrt{14}}{14}$. Therefore $\mathbf{r} \cdot \hat{\mathbf{n}} = \frac{5\sqrt{14}}{14}$.

10 a)

The unit normal is $\hat{\mathbf{n}} = \frac{1}{3\sqrt{5}}(-2, 4, 5)$. Since $A = (2, -3, 4)$,

$$A \cdot (-2, 4, 5) = -4 - 12 + 20 = 4,$$

so $\Pi_2 : \mathbf{r} \cdot \hat{\mathbf{n}} = \frac{4}{3\sqrt{5}} = \frac{4\sqrt{5}}{15}$.

10 b)

For Π_1, the constant is $\frac{8}{3\sqrt{5}} = \frac{8\sqrt{5}}{15}$. Thus

$$\Pi_1 : \mathbf{r} \cdot \hat{\mathbf{n}} = \frac{8\sqrt{5}}{15}, \qquad \Pi_2 : \mathbf{r} \cdot \hat{\mathbf{n}} = \frac{4\sqrt{5}}{15}.$$

10 c)

Both constants are positive, so the planes lie on the same side of the origin. Their separation, and hence the distance from A to Π_1, is

$$\frac{8\sqrt{5}}{15} - \frac{4\sqrt{5}}{15} = \frac{4\sqrt{5}}{15} \text{ units.}$$

11)

The plane normal is $(3, 9, 1)$. A perpendicular from $A = (2, -10, -642)$ meets the plane when $91\lambda - 728 = 0$, so $\lambda = 8$ and the foot is $(26, 62, -634)$. Hence

$$d = \sqrt{24^2 + 72^2 + 8^2} = 8\sqrt{91} \text{ units.}$$

Alternatively, $d = \frac{|3(2)+9(-10)-642-2|}{\sqrt{91}} = \frac{728}{\sqrt{91}} = 8\sqrt{91}$ units. A parallel-plane method gives the same distance: the plane through A parallel to Π has normalized constant $-726/\sqrt{91}$, while Π has constant $2/\sqrt{91}$, so their separation is $728/\sqrt{91}$.

12)

The direction vectors of the plane are $(3, 5, 7)$ and $(-2, 3, 9)$. A normal is

$$(3, 5, 7) \times (-2, 3, 9) = (24, -41, 19).$$

Using $(10, 66, -8)$ gives the plane equation $24x - 41y + 19z + 2618 = 0$. The perpendicular from the origin meets it at $\lambda(24, -41, 19)$, where $2618\lambda + 2618 = 0$, so the foot is $(-24, 41, -19)$. Therefore

$$d = \sqrt{24^2 + 41^2 + 19^2} = \sqrt{2618} \text{ units.}$$

Alternatively, $d = \frac{2618}{\sqrt{2618}} = \sqrt{2618}$ units.

13)

A normal to the plane is

$$(2,-5,1)\times(-1,1,3)=(-16,-7,-3).$$

Substituting $(120,-15,985)$ gives the plane equation $16x+7y+3z-4770=0$. The perpendicular from $(2,1,7)$ meets the plane at parameter $\lambda=15$ along direction $(16,7,3)$, at $(242,106,52)$. Thus

$$d=\sqrt{240^2+105^2+45^2}=15\sqrt{314}\text{ units.}$$

Alternatively, $d=\frac{4710}{\sqrt{314}}=15\sqrt{314}$ units.

14)

The two direction vectors in the plane are $(-1,2,4)$ and $(0,1,6)$. Their cross product is $(8,6,-1)$, so the plane has equation $8x+6y-z-39=0$, using $(2,4,1)$. The perpendicular from the origin meets the plane at $\lambda(8,6,-1)$, with $101\lambda-39=0$. Hence

$$d=\frac{39}{\sqrt{101}}=\frac{39\sqrt{101}}{101}\text{ units.}$$

15)

The planes have the same normal vector $(1,1,1)$. The distance is

$$\frac{|60-40|}{\sqrt{3}}=\frac{20\sqrt{3}}{3}\text{ units.}$$

16 a)

Substitution of the two points into $px+2y+qz=300$ gives

$$100p-4q=600,\qquad 150p+10q=500.$$

Solving gives $p=5$ and $q=-25$.

16 b)

The normals are $(5,2,-25)$. The plane constants in the form $5x+2y-25z+d=0$ are $d=-627$ for Π_2 and $d=-300$ for Π_1. Thus

$$d(\Pi_1,\Pi_2)=\frac{327}{\sqrt{654}}=\frac{\sqrt{654}}{2}\text{ m.}$$

17)

A normal to Π_1 is $(1,2,2)$, of length 3. A perpendicular displacement of length 5 is $\pm\frac{5}{3}(1,2,2)$. Starting from $(0,0,5)$ on Π_1 gives points $(\frac{5}{3},\frac{10}{3},\frac{25}{3})$ and $(-\frac{5}{3},-\frac{10}{3},\frac{5}{3})$ on Π_2. Substitution gives the two possibilities

$$\Pi_2: x+2y+2z-25=0\quad\text{or}\quad\Pi_2: x+2y+2z+5=0.$$

Alternatively, applying the point-plane distance formula to a parallel plane $x+2y+2z+d=0$ through $(0,0,5)$ gives $|10+d|/3=5$, so $d=5$ or $d=-25$.

18 a)

The line direction from $(4, 2, 0)$ to $(4, 10, 8)$ is proportional to $(0, 1, 1)$. The angle condition gives

$$\frac{12}{\sqrt{p^2 + 90}\sqrt{2}} = \frac{\sqrt{3}}{3},$$

so $p^2 = 126$. Since $p > 0$, $p = 3\sqrt{14}$.

18 b)

The laser has equation $\mathbf{r} = (4, 2, 0) + \lambda(0, 1, 1)$. Substitution into $3\sqrt{14}x + 9y + 3z = 6$ gives $\lambda = -1 - \sqrt{14}$, so the first point of contact is $(4, 1 - \sqrt{14}, -1 - \sqrt{14})$.

18 c)

The plane normal has length $\sqrt{(3\sqrt{14})^2 + 9^2 + 3^2} = 6\sqrt{6}$. The distance between the planes is $\frac{|26-6|}{6\sqrt{6}} = \frac{5\sqrt{6}}{9}$ m.

19)

The line direction $(3, 2, 5)$ is perpendicular to the plane normal $(-31, 14, 13)$ because their dot product is zero. The line is not contained in the plane: its point $(2, 8, -1)$ gives $-31(2) + 14(8) + 13(-1) + 2615 = 2652 \neq 0$. A perpendicular from this point meets the plane at parameter $\lambda = -2$, at $(64, -20, -27)$. Hence

$$d = \sqrt{62^2 + (-28)^2 + (-26)^2} = 2\sqrt{1326} \text{ units.}$$

Alternatively, $d = \frac{2652}{\sqrt{1326}} = 2\sqrt{1326}$ units.

20)

The line can be written $\mathbf{r} = (0, 3, 10) + \lambda(\frac{1}{2}, 2, -\frac{13}{6})$. Its direction is perpendicular to the plane normal $(3, -4, -3)$, so the line is parallel to the plane. The point $(0, 3, 10)$ is not on the plane. The perpendicular from it has direction $(3, -4, -3)$ and meets the plane at parameter $\lambda = 3$, at $(9, -9, 1)$. The distance is

$$\sqrt{9^2 + (-12)^2 + (-9)^2} = 3\sqrt{34} \text{ units.}$$

Alternatively, $d = \frac{102}{\sqrt{34}} = 3\sqrt{34}$ units.

21)

The distance from $P = (2, 1, 2)$ to the plane $4x + y + 8z + 65 = 0$ is

$$\frac{|8 + 1 + 16 + 65|}{\sqrt{4^2 + 1^2 + 8^2}} = 10.$$

Let $\mathbf{a} = (0, -13, 0)$ and $\mathbf{d} = (4, 3, k)$ describe the line. Equating the point-to-line distance to 10 gives

$$\frac{|\mathbf{d} \times (P - a)|}{|\mathbf{d}|} = 10, \qquad P - a = (2, 14, 2), \qquad \mathbf{d} \times (P - a) = (6 - 14k, 2k - 8, 50).$$

Squaring and simplifying gives $1 = \frac{2k^2 - 2k + 26}{25 + k^2}$, so $(k - 1)^2 = 0$ and $k = 1$.